

PRANA

Local-First AI Design Intelligence

COMPREHENSIVE SYSTEMS ARCHITECTURE & WORKFLOW REPORT

VERSION 1.0

ARCHITECTURE Next.js (Frontend) → FastAPI (Backend) → ComfyUI (Inference)

TARGET HARDWARE COMPUTE Nvidia RTX 5050 (8GB VRAM Profile)

1. Executive Summary

PRANA is a secure, local-first generative design workstation built to transform human intent into high-fidelity architectural, spatial, and typographic concepts. By bypassing cloud-compute dependencies, PRANA ensures zero-latency prompt orchestration, total data privacy, and hardware-optimized execution tailored specifically for an 8GB VRAM local environment.

This document outlines the system's core orchestration layers, from the React frontend to the ComfyUI inference engine, highlighting the graceful fallback mechanisms and deterministic prompt intelligence.

2. Core Execution Pipeline & Data Orchestration

The PRANA architecture utilizes a strictly decoupled client-server model. The frontend handles state and telemetry, while the FastAPI backend acts as the definitive source of truth, managing prompt intelligence, model routing, and database persistence.

The Inference Data Flow

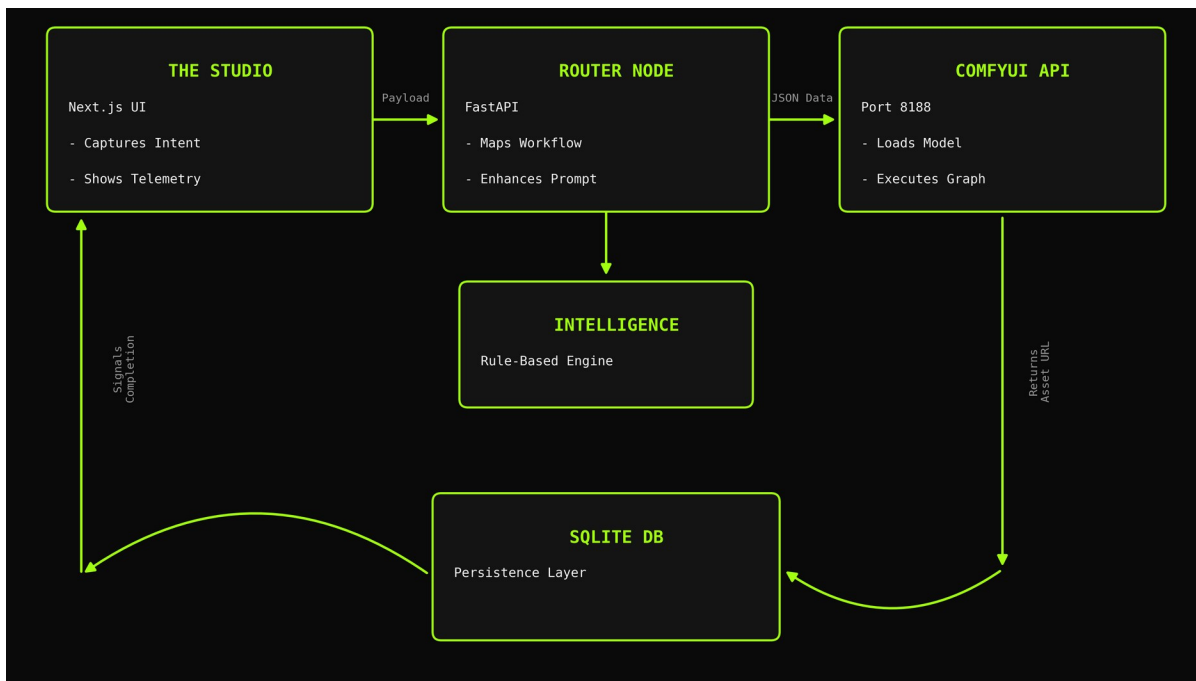


Figure 2.1 — End-to-end request flow: The Studio → Router Node → ComfyUI API, with the Intelligence layer and SQLite persistence layer completing the loop.

Component Breakdown

1. **Payload Ingestion:** The client sends a minimalist payload `{"prompt": "...", "mode": "campus-vision", "width": 768, "height": 768}`.

2. Intelligence Layer: The backend intercepts the payload. It does not send the raw prompt to the GPU. Instead, it processes it through the PromptIntelligence service, ensuring deterministic stylistic outputs.
3. Queue Orchestration: The payload is packed into a ComfyUI-compatible JSON graph mapping (linking nodes like KSampler and CLIPTextEncode) and pushed to the local 127.0.0.1:8188 cluster.

3. Creative Modes & Visual Output Specifications

The system isolates creative outputs into distinct operational modes. Below are the designated spaces for the generated visual assets, alongside the specific engineering prompts required to render them.

A. Campus Vision Pipeline

Purpose: Future-facing architectural layouts, spatial prototyping, and university infrastructure concepts.

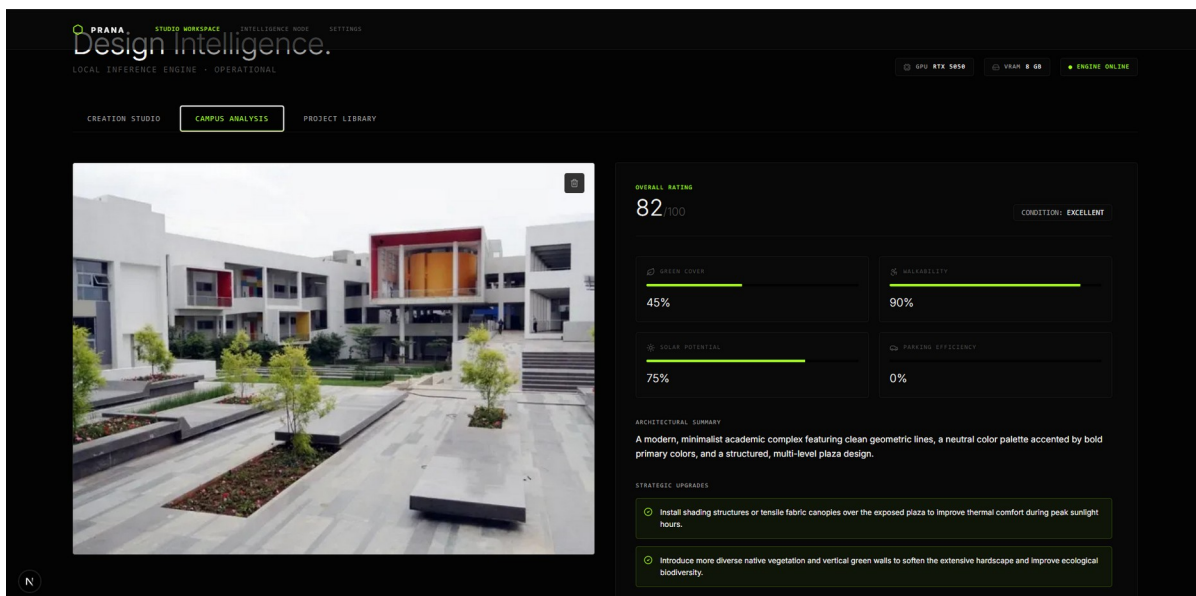


Figure 3.1 — The Campus Analysis module in the live PRANA workstation, scoring an uploaded courtyard photograph for green cover, walkability, and solar potential using the same design-intelligence layer that powers Campus Vision generation.

SYSTEM PROMPT TO INPUT FOR THIS IMAGE

A sweeping, photorealistic architectural visualization of a futuristic university campus plaza at twilight. Adaptive reuse of existing concrete structures integrated with seamless glass facades. Climate-responsive design featuring massive overhead solar canopies providing shaded circulation corridors. Warm, inviting volumetric lighting, highly detailed native vegetation and landscaping, students walking along pathways, 8k resolution, Unreal Engine 5 render style.

B. Poster Studio Pipeline

Purpose: Campaign systems, event marketing, and strict typographic hierarchy.

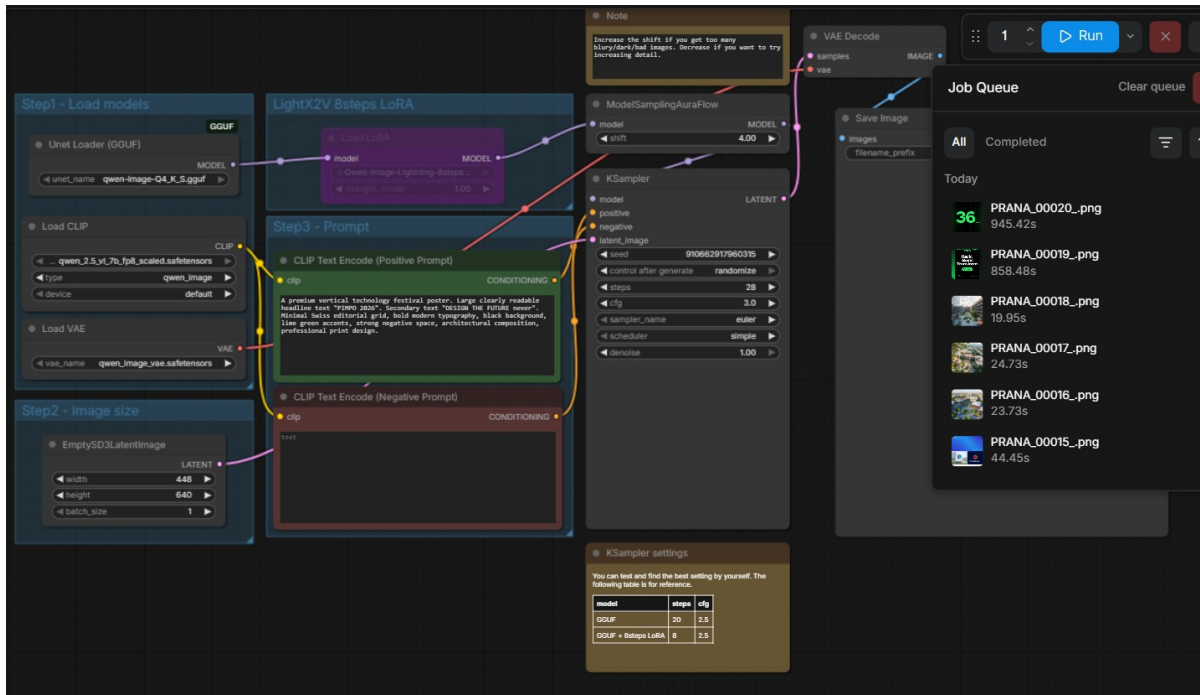


Figure 3.2 — Live ComfyUI execution graph for the Poster Studio pipeline: Qwen-Image GGUF checkpoint, LightX2V 8-step LoRA, and the encoded positive prompt, alongside a completed render queue.

SYSTEM PROMPT TO INPUT FOR THIS IMAGE

A premium, minimal technology festival poster. Massive, bold, readable headline text 'INNOVATE'. Secondary subtext 'Design the Future'. Swiss graphic design editorial grid layout. Deep cinematic black background with vibrant lime green (#A3FF12) architectural accents and structural wireframes. Intentional negative space, professional print quality, highly legible.

C. Brand Concept Pipeline

Purpose: Identity directions, physical packaging mockups, and corporate visual territories.



IMAGE PENDING

Brand Concept render not yet captured — add screenshot when available.

SYSTEM PROMPT TO INPUT FOR THIS IMAGE

High-end brand identity mockup for a cyber-security firm. A sleek, matte black presentation box alongside a metal business card and a minimalist ID badge. The logo is a subtle, geometric lime-green emblem. Professional studio lighting, sharp focus, incredibly detailed textures, cohesive corporate visual system, Behance portfolio style.

D. System Telemetry & Workstation Interface

Purpose: The operational command center for the human user.

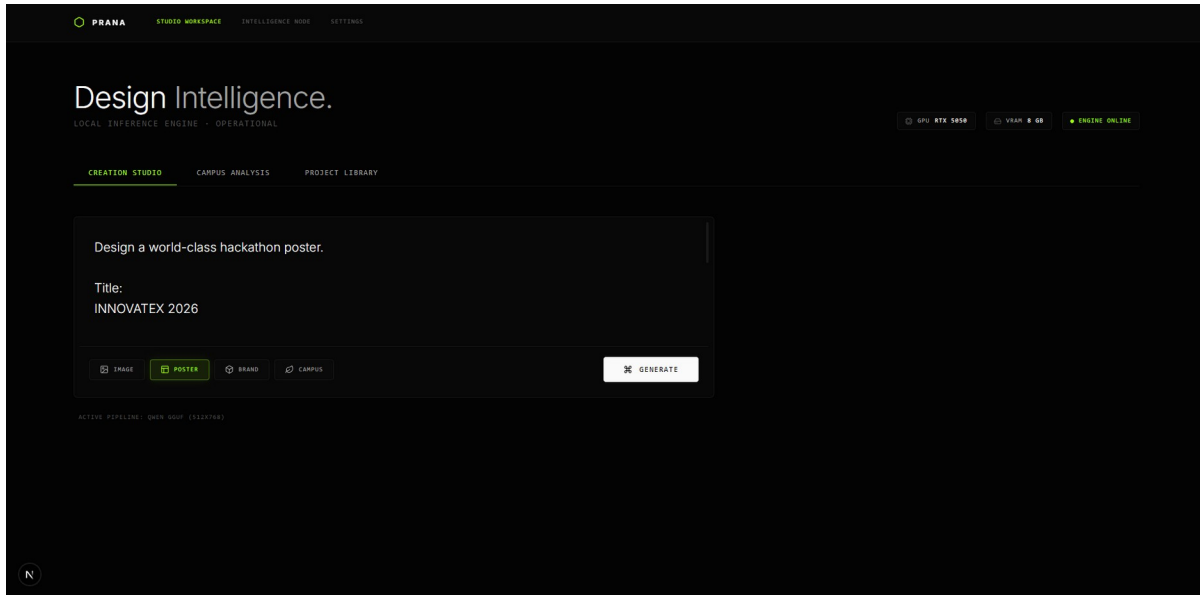


Figure 3.3 — The PRANA Creation Studio, showing the dark-mode workstation interface with live GPU (RTX 5050) and VRAM (8GB) telemetry, engine status, and the Poster mode prompt used to generate the “INNOVATEX 2026” hackathon poster.

SYSTEM PROMPT TO INPUT FOR THIS IMAGE

UI/UX design of a cinematic dark mode AI creative workstation, minimal architectural aesthetic, lime green accents, sleek data telemetry panels showing GPU and VRAM usage, monospace typography, high-end professional software interface, dribbble.

4. The Graceful Fallback Engine (Hardware Safeguard)

Operating complex generative models on an 8GB VRAM constraint requires strategic memory management. The PRANA Model Router is designed with a “Target vs. Fallback” configuration.

How the Router Evaluates Hardware

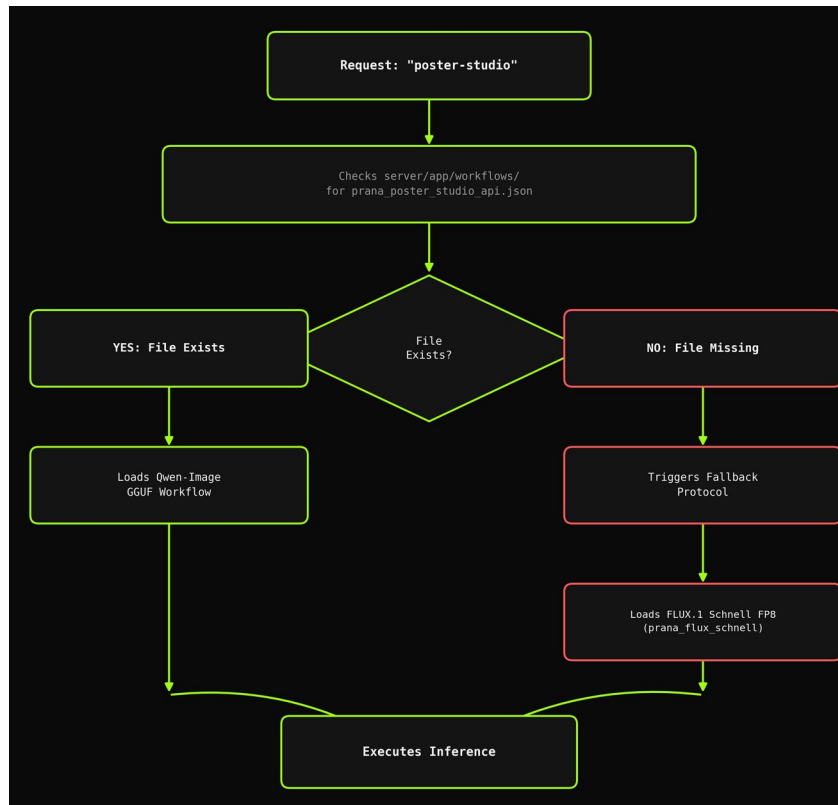


Figure 4.1 — Decision flow for the Model Router: a workflow-file existence check determines whether the request is routed to the target Qwen-Image pipeline or the FLUX.1 Schnell FP8 fallback.

Target State

If the user has downloaded a specialized checkpoint (e.g., a quantized Qwen-Image model for text rendering) and placed the corresponding JSON workflow in the server folder, PRANA routes the request directly to that specific local pipeline. Figure 3.2 above shows this pipeline live: the Qwen-Image GGUF checkpoint loaded alongside the LightX2V 8-step LoRA.

Fallback State

If the specialized model is absent, the backend refuses to crash. It logs a warning, intercepts the payload, and dynamically routes the enhanced prompt through the heavily verified FLUX.1 Schnell FP8 pipeline, recording the event in the SQLite database as “FLUX.1 Schnell FP8 (Fallback)” to maintain perfect data integrity.

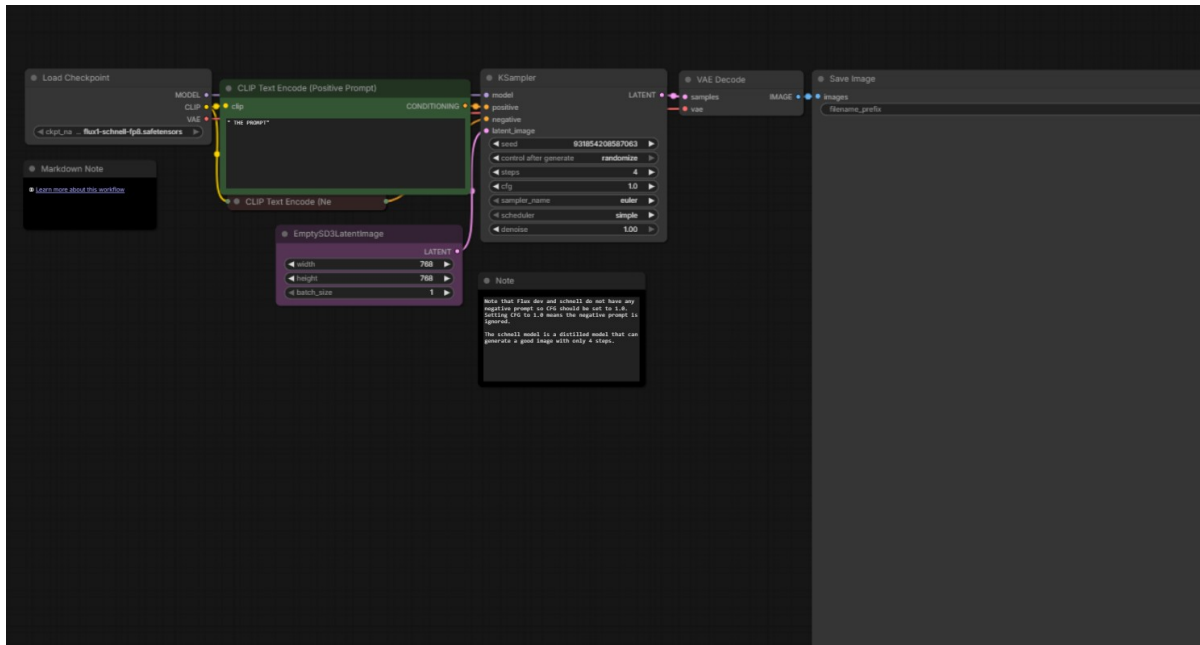


Figure 4.2 — Live ComfyUI execution graph for the Fallback State: the flux1-schnell-fp8 checkpoint running with CFG locked to 1.0 and only 4 sampling steps, per the distilled-model workflow notes.

5. Intelligence Layer: Deterministic Expansion

To ensure the output remains highly professional without requiring the user to type paragraphs of prompt engineering, PRANA utilizes a deterministic text-expansion layer.

The Logic Flow

4. User inputs: “A new library building”
5. User selects mode: Campus Vision
6. The PromptIntelligence.enhance() service intercepts this.
7. It evaluates the string for context clues. (e.g., If the user types “Bangalore”, it silently appends constraints for tropical savanna climate and monsoon-resilient structures).
8. It wraps the user's intent in a predefined architectural framework.

Final Effective Prompt Sent to GPU:

A new library building. Architectural visualization, photorealistic, adaptive reuse, preservation of existing structural elements, climate-responsive design, shaded circulation, passive cooling corridors, premium materials, high architectural precision.

This allows for rapid, iterative prototyping where the engine guarantees a baseline of artistic and technical quality on every generation cycle.